What Aspects of Earnings Calls Actually Predict Short-term Price Volatility?

Luke Brosnan

Economic & Social Research Institute, Dublin, Ireland

Abstract

This paper examines how earnings-call language predicts short-run post-call realised volatility. Using S&P 500 earnings-call transcripts matched to 5-minute intraday data, I construct event-time realised variance over 1 hour, 1 day, and 5 trading days after call completion and benchmark forecasts against HAR models augmented with SPY volatility lags and earnings-surprise controls. I show that transcript embeddings deliver large out-of-sample improvements in realised-volatility forecasts relative to HAR and dictionary-based text measures, with the strongest performance obtained using the full non-operator transcript. I then develop an interpretation framework that uses SHAP to identify the most important embedding components and links these latent semantic dimensions back to representative call sentences, and I introduce embedding-geometry features (breadth, traversal, novelty) computed from sentence-embedding sequences to assess whether structured summaries provide incremental predictive content beyond embedding averages, which are complementary to raw embedding vectors when forecasting volatility.

1. Introduction

Earnings conference calls are among the most information-dense corporate disclosure events. Beyond the mechanical release of earnings, calls combine prepared managerial narratives with unscripted exchanges in which analysts probe risks, guidance credibility, and operational constraints. A foundational literature establishes that earnings convey value-relevant information (Ball and Brown, 1968) and that markets may incorporate earnings news gradually, generating predictable post-announcement dynamics (Bernard and Thomas, 1989, 1990). Conference calls deepen this information channel by providing qualitative interpretation and forward-looking context. Early evidence shows that calls convey incremental information beyond the written release and are associated with sharp increases in trading activity and volatility during the call window (Tasker, 1998; Frankel, Johnson, and Skinner, 1999). Subsequent research emphasizes that information arrives through the internal structure of the call: both prepared remarks and Q&A are informative, with the unscripted analyst interaction often dominating the incremental information content (Matsumoto, Pronk, and Roelofsen, 2011). These features make earnings calls a natural setting for studying what aspects of disclosure language are most closely associated with short-run uncertainty and risk.

A parallel literature documents that the linguistic content of corporate disclosures predicts market outcomes beyond quantitative surprises. Studies using earnings-call tone and related textual measures find that language contains incremental information about market reactions, consistent with narratives conveying uncertainty, risk, and managerial credibility in ways not fully captured by earnings surprises alone (Price et al., 2012; Chen, Nagar, and Schoenfeld, 2018). Much of this work relies on dictionary-based measurement, which is attractive for transparency but limited by context insensitivity and an inability to capture semantic similarity across different wordings. Loughran and McDonald (2011) demonstrate that general-purpose sentiment dictionaries are ill-suited to financial text and propose finance-specific lexicons that substantially improve measurement in corporate disclosures. While lexicons remain widely used, they cannot represent negation, context, or the rich paraphrastic structure of language that is likely important in conference-call discourse.

Recent advances in natural language processing provide alternative representations. Transformer-based encoders generate contextual embeddings — continuous, high-dimensional representations of meaning, that can capture semantic nuance beyond word counts. Sentence embeddings, in particular, are designed to map semantically similar sentences close together in vector space (Reimers and Gurevych, 2019), enabling disclosure

content to be represented as dense numerical objects suitable for modern prediction methods. This shift aligns with a broader view in economics that text can be treated as high-dimensional data and incorporated into statistical and machine-learning frameworks (Gentzkow, Kelly, and Taddy, 2019). In finance, this has motivated a growing body of work using machine learning on text to improve prediction of returns and risk, particularly when disclosure language is measured using contextual representations rather than dictionaries.

The paper connects these strands to the realised-volatility forecasting literature. Realised volatility constructed from high-frequency returns is a precise and economically meaningful proxy for latent volatility (Andersen and Bollerslev, 1998; Andersen, Bollerslev, Diebold, and Labys, 2003). Its dynamics exhibit strong persistence and approximate long-memory behaviour, motivating benchmark forecasting models such as the Heterogeneous Autoregressive (HAR) specification (Corsi, 2009). Against this backdrop, the central question is whether earnings-call language contains incremental information for forecasting realised volatility measured in event time after the call concludes. The empirical design therefore evaluates nested forecasting specifications that begin with standard HAR predictors, add earnings-announcement controls, and then incorporate text features constructed either from finance-domain dictionaries or from transcript embeddings. Because embeddings are high-dimensional, the approach also considers low-dimensional representations (via principal component compression) to mitigate overfitting while preserving predictive content (Jolliffe, 2002).

Overall, the contribution is to provide evidence on which representations of earnings-call language, and which transcript segments, are most informative for short-horizon realised volatility forecasting. The focus on event-time horizons immediately after call completion targets the short-run uncertainty dynamics that are most plausibly linked to in-call information revelation, complementing existing work that emphasizes longer-run return predictability. The remainder of the paper describes the data and volatility construction, outlines the forecasting framework and evaluation design, and then develops interpretation and additional feature constructions that map the model's learned text dimensions back to economically meaningful content.

In line with expectations, specifications including embedding features massively outperform those including only dictionary, HAR and earnings features at short horizons, while any statistical difference between specifications decays by 5-days after a call has taken place. After identifying important embedding principal components and combinations of principal components in forecasting volatility, projection o these PCs back onto individual sentences

reveals the important topics/sentiments that explain volatility: XYZ. Each of the geometric features prove to be statistically significant predictors of post call realised variance, even when post call volatility is made orthogonal to embedding vector Principal Components. Addition of these geometric features to the non-parametric forecasting model yields a statistically significant improvement to out-of-sample forecast performance.

The remainder of the paper proceeds as follows. Section 3 describes the data and construction of text features and realized volatility. Section 4 outlines the forecasting framework, the interpretability layer and the construction and analysis of geometric features. Section 5 reports results and robustness, and Section 6 concludes.

2. Data

2.1. Earnings Call Transcripts

An earnings call is a quarterly conference call held by a publicly listed company, typically following the release of financial statements. During the call, senior executives discuss recent performance, provide forward-looking guidance, and comment on strategic developments, often followed by a Q&A session with analysts.

Earnings-call transcripts for S&P 500 firms from Q1 2019 to Q1 2026 were scraped from *the Motley Fool*, which publishes standardized company transcripts. To avoid survivorship bias, I construct a time-varying panel of S&P 500 constituents using a dynamically compiled list scraped from Wikipedia and maintained on GitHub[1], and retain only firms that are members of the index at the time of each call (and in the days immediately preceding it). The final dataset contains approximately XXXX transcripts for XXXX unique firms over a 7-year period.

Prior to constructing textual features, each transcript segment is cleaned using a standardized normalization pipeline to ensure consistent inputs across the corpus. The pipeline includes: (i) lowercasing to reduce spurious distinctions due to capitalization; (ii) light formatting normalization (standardizing whitespace and removing transcript artefacts such as repeated separators); and (iii) removal of non-linguistic characters when they do not convey semantic

[1] See repo:

content in this context. The output of this step is a cleaned sequence of text segments that serves as the standardized input for the embedding pipeline described below.

After cleaning, text is tokenised before being processed by the embedding model. A token is the basic unit the model reads. Tokenisation maps raw text into a sequence of token IDs, which allows the model to handle rare words and domain-specific terminology by splitting them into familiar subcomponents. These token sequences are then encoded by a transformer into contextual representations that are pooled into sentence-level embeddings.

[INSERT TABLE OF DESCRIPTIVE STATISTICS: TRANSCRIPT LENGTH/SPEAKERS/TURNS/QA/PREPARED REMARKS]

2.1.1. Dictionary-based text measures (Loughran–McDonald)

Before turning to embeddings, I also construct conventional dictionary-based measures of tone and risk language using the Loughran–McDonald (LM) financial dictionary (Loughran and McDonald, 2011). Dictionary methods map each transcript into counts (or shares) of words belonging to pre-defined semantic categories (e.g., positive, negative, uncertainty, litigious, constraining). Formally, let $T_{i,t}$ denote the cleaned transcript text for firm $i$at date t, tokenised into words $w \in \mathcal{V}$. For category $c \in \{\text{Positive}, \text{Negative}, \text{Uncertain}, \text{Litigious}, \text{Constraining}\}$ with dictionary $D_c \subset \mathcal{V}$, the category count is

$$\mathrm{LM}_{i,t}^{(c)} = \sum_{w \in T_{i,t}} \mathbb{1}\{w \in D_c\},$$

and I additionally compute transcript length and a net tone measure,

$$\mathrm{LM}_{i,t}^{(\text{net})} = \mathrm{LM}_{i,t}^{(\text{pos})} - \mathrm{LM}_{i,t}^{(\text{neg})}, \qquad \mathrm{WC}_{\,i,t} = \sum_{w \in T_{i,t}} 1.$$

These measures provide a transparent benchmark that has been widely used in empirical finance. However, dictionary approaches treat words as independent and ignore compositional meaning (e.g., negation, modality, and context), so semantically similar statements may be scored differently if they use different vocabulary. This limitation motivates the embedding-based representations described next, which encode meaning through the geometry of dense vectors rather than through discrete word lists.

[INSERT TIME SERIES GRAPH OF LM-COUNTS/SHARES HERE]

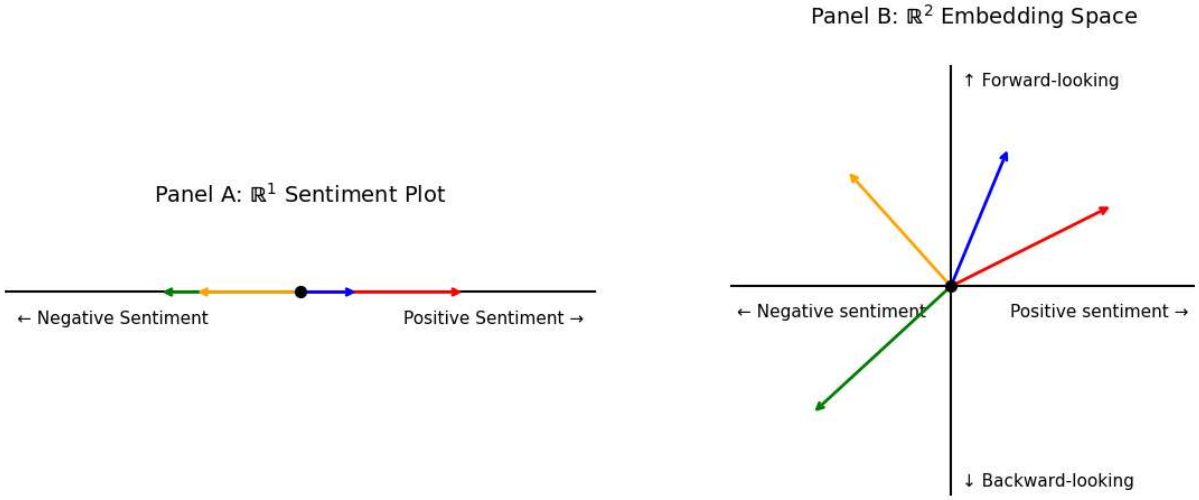
2.1.2. Large Language Models and Embedding Representations

Large language models (LLMs) represent text in a numerical form that supports semantic comparison. In practice, modern embedding models use a transformer encoder to map a sentence or document into a dense vector ("embedding") in a continuous, high-dimensional space. The encoder is trained so that texts with similar meaning have embeddings that lie close together (typically under cosine similarity), while dissimilar texts are farther apart. As a result, linguistically different sentences with similar meaning such as "We anticipate strong top-line growth" and "The firm expects revenues to increase" are proximate in embedding space despite having no overlapping vocabulary. This representation captures meaning, tone, temporal orientation, and contextual nuance in a way that bag-of-words approaches cannot.

Transformer encoders construct embeddings by contextualising each token in a sequence using self-attention: each token representation is updated based on its weighted interaction with other tokens in the same sentence. The final sentence embedding is then produced by pooling these contextual tokens, yielding a single fixed-length vector. Importantly, the geometry of the embedding space is learned from data: the model is trained so that semantically similar texts are pulled together, and unrelated texts are pushed apart. This training objective makes embeddings particularly well suited for measuring similarity and narrative structure in long-form communication such as earnings calls.

Although embeddings are high-dimensional, their geometry can be illustrated in low dimensions. Figure XX provides an intuition using synthetic vectors. Panel A depicts a one-dimensional space in which position corresponds to a sentiment-like direction (negative to positive). Panel B adds a second axis capturing temporal orientation (backward- vs forward-looking). These panels are illustrative: in real embedding models, individual coordinates are not labelled or directly interpretable as "sentiment" or "temporal focus". Instead, semantic attributes are encoded jointly across many dimensions, so meaning is reflected in the relative geometry of vectors through distances, angles, and directions, rather than in any single coordinate. The contrast between Panels A and B highlights this point, as two sentences may be similar along one attribute (e.g., both negative in sentiment) yet differ along another (e.g., backward- vs forward-looking), making them appear closer in $\mathbb{R}^1$ but farther apart once additional dimensions are considered.

Figure XX: Illustration of embedding-space geometry in low dimensions



In this sense, embedding-based measures generalise traditional one-dimensional sentiment scores by capturing multiple facets of language simultaneously, with those facets encoded in a correlated, high-dimensional representation. This can be extended to any number of semantic dimensions, creating a highly sophisticated and deeply understanding systems that can capture multiple features of a text. Relative position and direction encode both meaning and nuance, allowing downstream models to reason about similarity, emphasis, or rhetorical strategy in ways that traditional sentiment scores cannot capture.

Embeddings in this study are generated using $\mathrm{BAAI/bge\text{-}base\text{-}en\text{-}v1.5}$, a 768-dimensional encoder that achieved state-of-the-art performance on the 2023 Massive Text Embedding Benchmark (MTEB). The model is optimised for semantic similarity tasks, making it well suited for measuring the narrative and tonal structure of earnings-call communication.

Each transcript for firm i on date t is split into sentences $\{s_1, \dots, s_K\}$. Each sentence is then embedded via

$$\psi(s_k) \in \mathbb{R}^{768},$$

producing a matrix of sentence-level embeddings

$$Z_{i,t}^{\text{sentence}} = \begin{bmatrix} \psi(s_1) \\ \psi(s_2) \\ \vdots \\ \psi(s_K) \end{bmatrix} \in \mathbb{R}^{K \times 768}.$$

I then construct a call-level representation by mean pooling across sentences,

$$e_{i,t} = \frac{1}{K}\sum_{k=1}^{K} \psi(s_k) \in \mathbb{R}^{768},$$

which summarizes the transcript's central semantic content. This pooled embedding provides a low-noise representation of the overall sentiment, narrative and tone of the call and serves as the input for the downstream geometric measures used in the empirical analysis.

2.2. Financial data

Intraday 5-minute OHLVC data for S&P 500 tickers and the SPY ETF are obtained from financial modelling prep (FMP). Information on earnings including earnings announcement dates, actual earnings and expected earnings was also obtained from FMP. From these Intraday constituent bars, I construct realised volatility variables for 1- and 5-day horizons, which I present in detail in the two subsequent subsections.

2.2.1. Realised Volatility and its components

I construct measures of realised volatility using intraday returns. For each ticker i and trading period t let

$$r_{i,t,j} = p_{i,t,j} - p_{i,t,j-1}, \qquad j = 1,2,\dots,J$$

denote the 5-minute stock return of ticker i at the j-th interval of period t, where J represents the total number of 5-minute intervals in the period.

The baseline realised volatility measure over the period t is then the sum of squared intraday returns over that period,

$$RV_{i,t} = \sum_{j=1}^{J} r_{i,t,j}^2$$

For multi-day horizons, I compute cumulative realised measures over H consecutive trading days for all realised volatility outcomes. Fo instance, $RV_{i,t}^{(H)}$ represents the cumulative realised volatility observed from day $t - H + 1$ up to day t,

$$RV_{i,t}^{(H)} = \sum_{h=0}^{H-1} RV_{i,t-h}$$

2.2.2. Earnings surprises

Earnings surprises capture the extent to which reported earnings deviate from market expectations and are a key driver of post-announcement price reactions. I compute two standard measures for each event. First, the standardised earnings surprise (SUE) is defined as

$$SUE_{i,q} = \frac{EPS_{i,q} - \mathbb{E}[EPS_{i,q}]}{\sigma(EPS_i)}$$

where $EPS_{i,q}$ is the reported earnings per share of firm i in quarter q, $\mathbb{E}[EPS_{i,q}]$ is the consensus analyst forecast immediately prior to the announcement, and $\sigma(EPS_i)$ is the historical standard deviation of forecast errors for firm i[2]. This scaling makes the surprise comparable across firms and time.

Second, I compute the earnings surprise ratio (ESR),

$$ESR_{i,q} = \frac{EPS_{i,q} - \mathbb{E}[EPS_{i,q}]}{\left|\mathbb{E}[EPS_{i,q}]\right|}$$

which measures the proportional deviation of reported earnings from consensus expectations.

Both measures are constructed separately for quarterly announcements and are merged with the corresponding transcript data. By incorporating standardized and proportional surprises, the framework can capture both the magnitude and the relative importance of unexpected earnings news when assessing predictive performance[3].

3. Methodology

3.1. Forecast design

All forecasting models are estimated using LightGBM, an efficient implementation of gradient-boosted decision trees (Friedman, 2001; Ke et al., 2017). For an outcome y_i and predictors x_i, gradient boosting constructs an additive predictor of the form

$$\hat{g}(x) = \sum_{m=0}^{M} \eta\, T_m(x),$$

[2] $\sigma(EPS_i)$ is calculated using earnings info from a minimum of x consecutive quarters and a maximum of y consecutive quarters?

[3] Something about other earnings variables I use/create

where $T_m(\cdot)$ denotes a shallow regression tree, $\eta \in (0,1]$ is the learning rate, and M is the number of boosting iterations. Starting from a constant baseline,

$$\hat{g}_0 = \arg\min_c \sum_{i=1}^{n} L(y_i, c),$$

the algorithm proceeds iteratively by computing pseudo-residuals

$$r_i^{(m)} = -\frac{\partial L(y_i, g)}{\partial g} \Big|_{g=\hat{g}_{m-1}(x_i)},$$

fitting a regression tree to these residuals, and updating the predictor according to

$$\hat{g}_m(x) = \hat{g}_{m-1}(x) + \eta T_m(x).$$

LightGBM preserves this boosting framework while using histogram-based splitting and leaf-wise tree growth, which substantially improves computational efficiency in settings with high-dimensional predictors such as transcript embeddings.

The forecasting exercise is conducted in event time around earnings calls. The dependent variables are horizon-specific realized volatility measures constructed from 5-minute intraday returns after the call, transformed as $\log(1 + RV)$ (Andersen and Bollerslev, 1998; Andersen et al., 2003). The empirical implementation considers three post-event horizons: one hour, one trading day, and five trading days. For each earnings call, the volatility window begins at the call finish time whenever the call start time and duration are observed. If duration is missing, a duration of 60 minutes is assumed. If the call time is unavailable, the post-event window begins at the next trading-day open. This design ensures that realized volatility is measured strictly after the information release embodied in the call.

For each event t and forecast horizon h, the target variable is modelled as

$$y_{t,h} = F_\theta(\mathcal{X}_t) + \varepsilon_{t,h},$$

where $F_\theta(\cdot)$ is approximated by LightGBM and $\mathcal{X}_t$ is a set of structured predictor blocks. The baseline block follows the heterogeneous autoregressive (HAR) approach and consists of lagged firm-level realized volatility at the 1-day, 5-day, and 21-day horizons, all measured in logs and shifted by one day to preserve forecasting validity. To account for market-wide volatility conditions, an analogous set of HAR variables is also constructed for the S&P 500 ETF (SPY) and merged by date. The resulting baseline therefore incorporates both firm-specific and aggregate market volatility information.

The baseline is then augmented using progressively richer information sets. The first extension adds earnings-announcement variables retained from the event-level data, including earnings surprises and timing-related controls. The second additionally incorporates Loughran–McDonald dictionary measures computed from the relevant call text. The third introduces a dense semantic representation by augmenting the HAR and earnings blocks with an embedding-based transcript representation, constructed as the mean of sentence embeddings within the relevant text sample.

All text-based specifications are estimated across multiple transcript samples that vary along two dimensions: whose speech is included and whether the text is pooled across the full call or separated into prepared and Q&A components. Specifically, the analysis considers (i) the full call excluding operator speech, either as a single pooled transcript or with prepared remarks and Q&A treated as distinct segments; (ii) management speech only, again either pooled across the call or segmented into prepared and Q&A; (iii) management speech in the Q&A segment only; and (iv) prepared remarks only. This design allows a direct assessment of whether predictive content resides primarily in management versus analyst contributions, and whether it is concentrated in the prepared portion of the call or in the interactive Q&A, while keeping the forecasting and tuning protocol otherwise identical across text samples.

Because transcript embeddings are high-dimensional, the embedding block is compressed using principal components analysis prior to estimation. Let $Z \in \mathbb{R}^{N \times d}$ denote the matrix of transcript-level embeddings, where d is the embedding dimension. The reduced representation is given by

$$Z_{\text{PCA}}^{(k)} = ZW_k,$$

where $W_k \in \mathbb{R}^{d \times k}$ contains the leading eigenvectors of the covariance structure of Z. In the empirical implementation, $k \in \{10,50,100\}$. This dimensionality reduction mitigates overfitting risk when a relatively large embedding space is mapped into a finite event sample. Prior to PCA, numeric predictors are standardised and missing values are imputed using simple deterministic rules, so that the preprocessing pipeline is consistent across estimation and interpretation.

Model evaluation follows a strict chronological design based on actual call dates. The full set of unique event dates is ordered in time, and the final 20 per cent of dates is reserved as a holdout test sample. Of the remaining pre-test dates, the final 20 per cent are used as a validation block, with all earlier dates assigned to the training sample. This design preserves the temporal

structure of the forecasting problem and avoids contamination from future information. Hyperparameters are selected using the validation block only. For each specification, the procedure searches over a grid of LightGBM parameters governing tree complexity, subsampling, and L_1/L_2regularisation, while also selecting the embedding dimension k when PCA is employed. Early stopping is implemented using a further time-ordered holdout taken from the tail of the training sample, thereby controlling overfitting without violating chronological ordering. Once the preferred hyperparameter configuration is identified, the final model is re-estimated on the combined training and validation data and then evaluated on the holdout test sample.

Forecast performance is assessed using out-of-sample mean squared error, root mean squared error, and R^2. To compare nested specifications formally, Diebold–Mariano tests are applied to out-of-sample forecast errors relative to the HAR benchmark (Diebold and Mariano, 1995). For multi-day horizons, heteroskedasticity and autocorrelation consistent standard errors are used in order to account for error dependence induced by overlapping forecast windows. Additional residual diagnostics are also examined, including Ljung–Box tests and simple tests of residual predictability (Ljung and Box, 1978), in order to verify that improvements in predictive accuracy are not merely reflecting residual serial dependence left unmodelled by the forecasting system.

3.2. Embedding Interpretability

Interpretability is based on Shapley Additive Explanations (SHAP), which provide an observation-level decomposition of a fitted machine-learning prediction into additive feature contributions (Shapley, 1953).

The SHAP framework is derived from the Shapley value concept in cooperative game theory. In that setting, each feature is viewed as a "player" in a coalition, and the predicted value for a given observation is the outcome generated by the joint contribution of all players. The Shapley value for a feature measures its average marginal contribution to the prediction across all possible subsets of the remaining features (Lundberg and Lee, 2017). Formally, let $f(x_i)$ denote the fitted model prediction for observation i, and let $\phi_{i,k}$ denote the SHAP value for feature k. SHAP represents the prediction as

$$f(x_i) = \phi_{i,0} + \sum_{j=1}^{p} \phi_{i,k},$$

where $\phi_{i,0}$ is a baseline term, typically the average model prediction, and $\phi_{i,k}$ is the contribution of feature k to the deviation of observation i's prediction from that baseline. Hence, if $\phi_{i,k} > 0$, feature $j$raises the predicted volatility for observation $i$relative to the baseline; if $\phi_{i,k} < 0$, it lowers it. This decomposition is particularly useful in nonlinear models such as boosted trees, where the effect of a feature cannot be read directly from a single regression coefficient and may depend on interactions with many other variables.

The Shapley contribution for feature $j$can be written as

$$\phi_k = \sum_{S \subseteq K \setminus \{k\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \left[v(S \cup \{k\}) - v(S)\right],$$

where K is the full set of features, S indexes a subset excluding k, and $v(S)$ is the model value when only the subset S is included. The term in brackets measures the marginal contribution of feature k when added to subset S, and the combinatorial weight ensures that all possible feature orderings are treated symmetrically. In tree-based models, efficient algorithms exist for computing these quantities, making SHAP feasible in high-dimensional settings (Lundberg et al., 2020).

In the present setting, SHAP is computed on the transformed test-set design matrix produced by the fitted preprocessing pipeline. Contributions are therefore defined with respect to the variables that actually enter the LightGBM model, including HAR predictors, earnings-announcement controls, and, crucially, the principal components of the transcript embedding block. A global importance measure is constructed by averaging absolute SHAP values across the test sample:

$$I_k = \frac{1}{N} \sum_{i=1}^{N} |\phi_{i,k}|.$$

This statistic captures the average magnitude of a feature's contribution, irrespective of sign, and provides a natural ranking of the variables the model relies upon most heavily.

The analysis then focuses specifically on the embedding principal components (PCs) used by the model. Let $PC_{i,\ell}$ denote observation i's score on embedding principal component ℓ, and let $\phi_{i,\ell}$ denote the corresponding SHAP value. For each component, its average importance is measured by

$$I_\ell = \frac{1}{N}\sum_{i=1}^{N}\left|\phi_{i,\ell}\right|.$$

To recover direction, I examine the empirical association between the PC score and its SHAP contribution:

$$s_\ell = \text{sign}\ (\text{corr}\ (PC_{i,\ell}, \phi_{i,\ell})),$$

and define a signed importance weight

$$w_\ell = I_\ell s_\ell.$$

The intuition is that I_ℓ identifies which semantic components matter for prediction, while s_ℓ indicates whether larger values of that component tend, on average, to push predicted volatility upward or downward. A component with $w_\ell > 0$ is therefore one that the model both uses strongly and associates with higher predicted post-call volatility; $w_\ell < 0$indicates association with lower predicted volatility.

Rather than mapping these document-level predictors back to individual sentences via a single composite projection, interpretation is conducted at the level of the embedding PCs themselves, which corresponds directly to the scale at which the forecasting model is estimated. Let $\bar{e}_i \in \mathbb{R}^{768}$ denote the transcript-level mean embedding for call $i$after the same standardisation used in estimation, and let $v_\ell \in \mathbb{R}^{768}$ denote the loading vector for principal component ℓ. Then

$$PC_{i,\ell} = \langle \bar{e}_i, v_\ell \rangle.$$

For each of the most important PCs, I characterise the associated semantic content by examining the sentences with extreme values of $PC_{i,\ell}$ (and, where relevant, by examining extreme combinations of pairs of PCs). Because these excerpts are selected conditional on the model's actual embedding representation, they are best interpreted as indicative of the linguistic content that places a transcript in a region of embedding space the model treats as more or less volatility-relevant, rather than as standalone "sentence effects."

This PC-level interpretation can be extended to interaction structure by examining pairs of high-importance components jointly. In particular, considering the two-dimensional space $(PC_{i,\ell}, PC_{i,m})$ allows the identification of transcript excerpts characteristic of high–high, high–low, low–high, and low–low regions, providing a transparent way to describe semantic interaction patterns that a nonlinear model such as gradient boosting may exploit. Together, these procedures provide an interpretable bridge between the fitted forecasting model and the

content of earnings calls while maintaining fidelity to the level at which text enters the prediction problem.

3.3. Geometric Features

To complement the forecasting models, I construct a set of transcript-level geometric features from sentence embeddings. These measures are designed to capture the internal semantic structure of each earnings call, including its breadth, semantic movement, and its novelty relative to recent calls. The feature-construction procedure begins from sentence-level embeddings $e_{i,s} \in \mathbb{R}^{768}$, where i indexes the earnings call, and s indexes the sentence within the call. Sentences are assigned to transcript segments using section labels, distinguishing prepared remarks and Q&A, and to speaker roles, distinguishing management, analysts, and operator speech. All features are then aggregated to the call level.

For any subset of sentences S_i within call i, the corresponding centroid is defined as

$$\bar{e}_i = \frac{1}{|S_i|} \sum_{s \in S_i} e_{i,s}.$$

In the implementation, centroids are constructed for the full document, for Q&A management speech, and for Q&A analyst speech, and are used both directly and as inputs to higher-order features.

3.3.1. Semantic Breadth

Semantic breadth is measured as the Euclidean norm of the per-dimension standard deviations of sentence embeddings within a given segment. Let

$$\sigma_{i,d} = \sqrt{\frac{1}{|S_i|} \sum_{s \in S_i} \left(e_{i,s,d} - \bar{e}_{i,d}\right)^2},$$

where $d = 1, \ldots, 768$ indexes embedding dimensions. Raw breadth is then

$$B_i = \|\sigma_i\|_2 = \left(\sum_{d=1}^{768} {\sigma_{i,d}}^2 \right)^{1/2}$$

This is computed separately for the full transcript, prepared remarks, and Q&A. As raw breadth is mechanically related to transcript length, it is orthogonalized with respect to log sentence count. Specifically, for each segment S, the following auxiliary regression is estimated across calls:

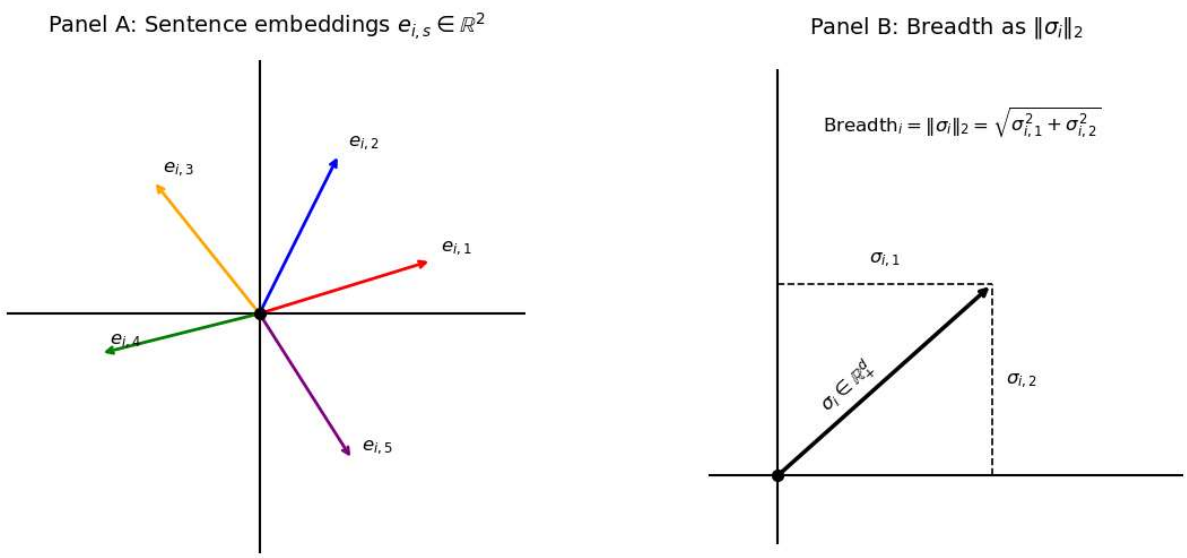
$$B_i = \alpha + \beta \log(1 + n_i) + u_i,$$

where n_i is the number of sentences in the relevant segment. The residual

$$\text{Breadth}_i = B_i^{\perp} = u_i$$

is the length-adjusted breadth measure used in the empirical analysis.

Figure XX: Graphical representation of Breadth



3.3.2. Semantic Traversal

Traversal measures how much the transcript moves through semantic space as it unfolds. Let the sentences within segment S_i be ordered as $e_{i,1}, \dots, e_{i,n_i}$. Each sentence vector is first normalised to unit length,

$$\tilde{e}_{i,s} = \frac{e_{i,s}}{\|e_{i,s}\|}.$$

Traversal is then based on cosine distance between consecutive sentences. For any adjacent pair,

$$\delta_{i,s} = \cos(e_{i,s}, e_{i,s-1}) = 1 - \tilde{e}_{i,s}{}^{\top} \tilde{e}_{i,s-1}.$$

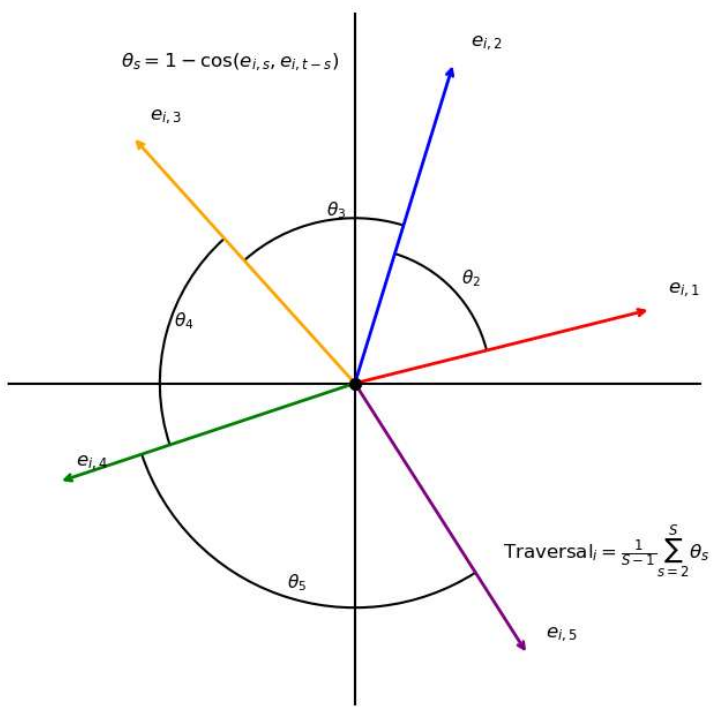
For a relevant subset of sentences S_i, traversal is then the average cosine distance between consecutive sentences,

$$\text{Trasversal}_i = \frac{1}{n_i - 1} \sum_{s=2}^{n_i} \delta_{i,s}.$$

This is constructed for the full document, prepared remarks, and Q&A. A high traversal value indicates that consecutive sentences tend to move more sharply across semantic space, while a low value indicates a more locally coherent semantic path.

This is constructed for the full document, prepared remarks, and Q&A. A high traversal value indicates that consecutive sentences tend to move more sharply across semantic space, while a low value indicates a more locally coherent semantic path.

Figure XX: Graphical representation of Traversal



3.3.3. Semantic Novelty

Novelty is constructed at the document level using a global rolling Ledoit–Wolf Mahalanobis distance (Mahalanobis, 1936; Ledoit and Wolf, 2004). First, each transcript is represented by its document centroid $\bar{e}_i$. For each call date t_i and a rolling calendar window of length $w \in \{21,63\}$ days, define the historical set

$$\mathcal{H}_{i,w} = \{j \mid t_i - w \leq t_j < t_i\},$$

that is, all prior calls occurring within the previous w calendar days.

For each window, let $\hat{\mu}_{i,w}$ and $\hat{\Sigma}_{i,w}$ denote the Ledoit–Wolf shrinkage estimates of the mean vector and covariance matrix of $\{z_j \mid j \in \mathcal{H}_{i,w}\}$. The novelty of call i (separately for $w \in \{21,63\}$) is then defined as the Mahalanobis distance

$$N_{i,w} = \sqrt{(z_i - \hat{\mu}_{i,w})' \hat{\Sigma}_{i,w}^{-1} (z_i - \hat{\mu}_{i,w})}.$$

Intuitively, these variables measure how unusual the semantic content of the current call is relative to the distribution of all prior calls in the recent global history.

Two additional relative geometry variables compare the Q&A with prepared remarks. The first is the ratio of $\mathrm{Trasversal}_i$ in call i's Q&A section and prepared remarks section, and the second is the additive difference in $\mathrm{Breadth}_i$ between call i's Q&A and prepared remarks.

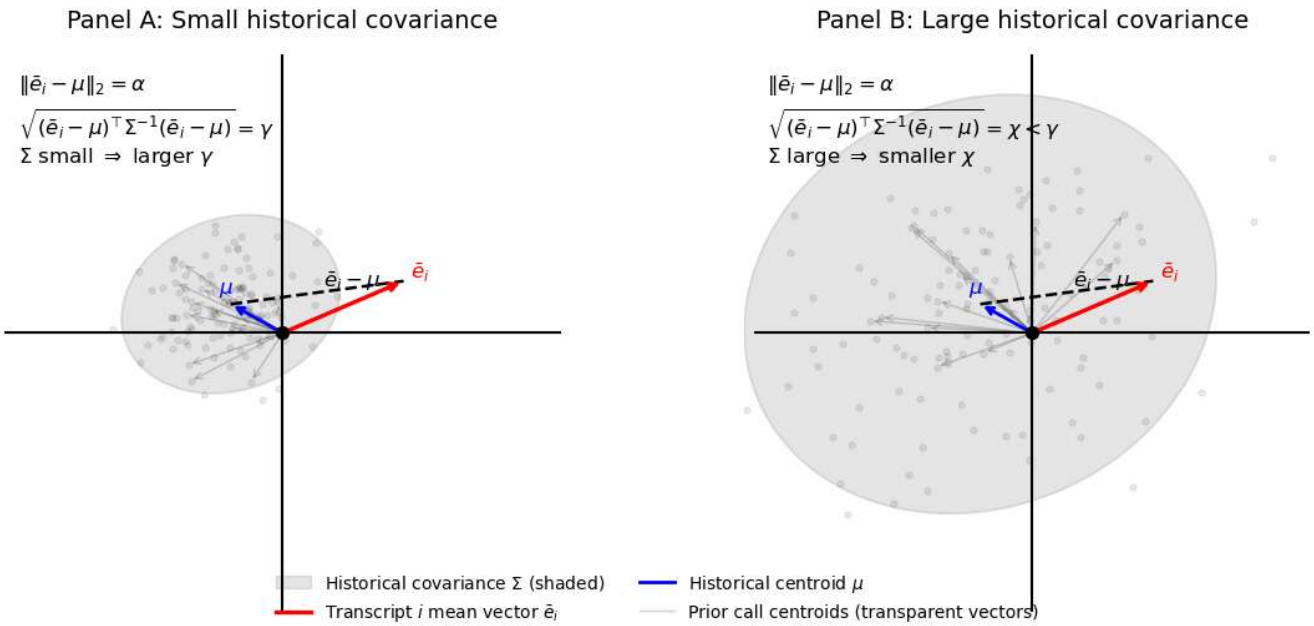
3.3.4. Regression Framework

The regression analysis relates post-call realised volatility to geometric transcript features in a panel setting. For call i and horizon h, the baseline specification is

$$y_{i,h} = \alpha + X_i^\top \beta + G_i^\top \gamma + \varphi + \tau + \varepsilon_{i,h},$$

where $y_{i,h} = \log(1 + \mathrm{RV}_{i,h})$ is realised volatility measured over the post-call window, X_i contains the benchmark volatility and earnings controls, G_i is a vector of geometric transcript features, φ denotes industry fixed effects τ denotes calendar month dummies. The control set X_i includes firm-level HAR predictors, SPY HAR predictors and earnings-announcement variables. All right-hand-side variables are standardised prior to estimation so that coefficients may be interpreted as the change in $y_{i,h}$ associated with a one-standard-deviation increase in the corresponding regressor.

Figure XX: Graphical representation of novelty



The geometric feature set is entered either one-by-one, in which case G_i contains a single geometric variable to quantify its marginal association with subsequent volatility, or as blocks, in which case G_i contains multiple geometry measures simultaneously. Because document novelty is constructed using both 21-day and 63-day rolling windows, all combined specifications are estimated separately under each novelty definition. In addition to the full block, reduced and document-only blocks are also estimated; the document-only specifications include only document novelty, document breadth, and document traversal, allowing a direct assessment of whether explanatory power is driven primarily by document-level semantic structure or by richer interaction-related features of the Q&A.

The preferred standard errors are clustered at the firm-level, and, where feasible, two-way clustering by firm and event date is also reported to accommodate both within-firm serial dependence and cross-sectional correlation within earnings-date clusters. For horizons with overlapping realised-volatility windows, HAC/Newey–West adjustments remain relevant (Newey and West, 1986)t; accordingly, HAC-based inference is reported as an additional robustness check, with lag length chosen to match the horizon (e.g., larger lag choice for longer overlapping windows). The emphasis is on demonstrating that the sign and magnitude of the geometry coefficients are robust across plausible inference choices.

A complementary set of regressions evaluates whether geometric features explain variation in realised volatility beyond what is already captured by the forecasting framework. Let $\hat{y}_{i,h}$ denote the out-of-sample prediction from the best baseline forecasting model that includes embeddings from section 4.1. Define the corresponding out-of-sample residual

$$\hat{\varepsilon}_{i,h} = y_{i,h} - \hat{y}_{i,h}.$$

I then estimate residual regressions of the form

$$\hat{\varepsilon}_{i,h} = \alpha + X_i^\top \beta + G_i^\top \gamma + \varphi + \tau + u_{i,h},$$

with the same clustering strategy as above. These regressions test whether the geometric variables capture systematic, predictable variation in realised volatility that is not already absorbed by the benchmark forecasting model's information set. Interpreted conservatively, significant geometry coefficients in these residual regressions indicate that geometric features provide incremental structure relative to the baseline model, rather than merely re-expressing information already captured by HAR, earnings controls and by embeddings.

3.3.5. Extended Forecast Framework

Finally, the nonparametric forecasting framework is extended to evaluate the incremental predictive content of geometric transcript variables, both in isolation and alongside embeddings. The forecasting model is estimated using the same LightGBM procedure, identical target construction, and the same strict chronological train/validation/test design described earlier. The benchmark specifications are augmented in two steps.

First, geometric features are added to the HAR and earnings-information baseline to form the specification

$$y_{i,h} = F_\theta\,(X_i, G_i) + \varepsilon_{i,h},$$

This specification evaluates whether the geometry block improves out-of-sample forecast accuracy relative to the HAR+EA benchmark.

Second, geometric features are added jointly with embeddings to form

$$y_{i,h} = F_\theta\,(X_i, Z_i, G_i) + \varepsilon_{i,h},$$

Comparing performance across $F_\theta(X_i, Z_i)$ and $F_\theta(X_i, Z_i, G_i)$ provides a direct test of whether the geometry variables contain incremental predictive information beyond raw semantic embeddings. If the geometry block does not improve predictive accuracy once embeddings are

included, this suggests that the geometric measures largely act as compressions or reparameterisations of information already present in the embedding vectors. Conversely, incremental gains from adding geometry on top of embeddings would indicate that these measures capture complementary structure that is not fully preserved by the document-level mean embedding representation.

Forecast comparison follows the same evaluation protocol as the main forecasting results: hyperparameters are selected on the validation block only, final models are re-estimated on the combined training and validation sample, and performance is assessed on the held-out test set using out-of-sample loss metrics and formal forecast comparison tests relative to the benchmark specifications.

4. Results

4.1. Baseline Forecast Results

This section summarises out-of-sample forecasting performance for realised variance across horizons. Performance is evaluated on the held-out test period using mean squared error (MSE) for $\log(1 + \mathrm{RV})$. For each horizon, percentage improvements are reported relative to the HAR benchmark, and statistical differences in forecast accuracy are assessed using Diebold–Mariano tests against HAR. Table XX reports results for the full-transcript-less-operator text universe; corresponding results for alternative text samples are provided in the appendix. In addition, Table XX reports the out-of-sample R^2 and the Ljung–Box p-value from residual diagnostics, which provide a summary check on whether forecast errors exhibit remaining serial correlation and whether improvements are accompanied by cleaner residual behaviour.

Across horizons, incorporating earnings information and textual features improves predictive performance most clearly at short horizons. At the 1-hour horizon, augmenting HAR with earnings variables reduces test MSE by 5.1% relative to the benchmark, while adding Loughran–McDonald dictionary measures does not yield further gains and slightly worsens performance. The main result is that embedding representations deliver large improvements in short-horizon realised-variance forecasts: relative to HAR, including document-level embeddings reduces test MSE by approximately 21%, with a commensurate increase in out-of-sample R^2. The same pattern holds at the 1-day horizon, where embeddings reduce MSE by roughly 22% and substantially increase out-of-sample fit compared to HAR and HAR+EA. In both horizons, the Diebold–Mariano p-values indicate that embedding-augmented specifications outperform the

HAR benchmark at conventional significance levels, consistent with embeddings capturing economically meaningful information beyond volatility persistence and earnings surprises.

Table XX: Realised variance forecasts

Horizon	Specification	MSE	% Improvement	R-squared	Ljung-Box p-value
1-hour	HAR	0.494	*Ref.*	0.197	0.10
	+Earnings	0.469	5.1%**	0.237	0.12
	+Earnings+Dictionary	0.471	4.6%*	0.233	0.14
	+Earnings+Document Embeddings	0.392	20.6%***	0.362	0.22
	+Earnings+Section Embeddings	0.39	21.0%**	0.365	0.39
1-day	HAR	0.492	*Ref.*	0.293	0.37
	+Earnings	0.477	3.10%	0.315	0.44
	+Earnings+Dictionary	0.474	3.70%	0.319	0.51
	+Earnings+Document Embeddings	0.382	22.5%**	0.452	0.50
	+Earnings+Section Embeddings	0.386	21.7%*	0.446	0.74
5-day	HAR	0.305	*Ref.*	0.496	0.46
	+Earnings	0.304	0.30%	0.497	0.36
	+Earnings+Dictionary	0.306	-0.30%	0.494	0.42
	+Earnings+Document Embeddings	0.293	4.00%	0.516	0.47
	+Earnings+Section Embeddings	0.297	2.90%	0.510	0.40

At the 5-day horizon, improvements are smaller. Embedding-augmented models reduce MSE modestly relative to HAR (on the order of a few percent), and while point estimates continue to favour embeddings, the Diebold–Mariano tests do not reject equal predictive accuracy at conventional thresholds. This attenuation is consistent with the broader finding that the incremental information in earnings-call language is most valuable for near-term volatility dynamics, whereas longer-horizon realised variance is dominated by more persistent volatility components already well captured by HAR-style predictors.

Allowing distinct embedding representations for prepared remarks and Q&A does not materially improve out-of-sample performance relative to a single document-level embedding. In several cases, the section-split specification performs similarly in MSE terms but yields weaker Diebold–Mariano evidence against the HAR benchmark, which is consistent with additional flexibility increasing estimation variance without delivering commensurate signal. Residual diagnostics provide a complementary perspective. Ljung–Box p-values are generally not

indicative of severe residual autocorrelation in the best-performing embedding specifications, suggesting that gains are not merely reflecting unmodelled serial dependence in the forecast errors. The out-of-sample R^2 results mirror the MSE patterns, reinforcing the conclusion that embeddings meaningfully improve short-horizon predictive performance.

Finally, results are qualitatively similar across alternative transcript universes, but restricting the text to management-only speech or to a single call segment (prepared remarks only or Q&A only) tends to reduce forecast performance slightly relative to the full transcript. This suggests that the predictive content relevant for post-call realised volatility is distributed across the broader earnings-call interaction rather than being confined to a single speaker group or segment, consistent with market participants processing the full informational environment of the event.

Although embedding-enhanced specifications outperform the HAR benchmark on average even at the 5-day horizon, the differences are not statistically distinguishable from zero according to the Diebold–Mariano tests, and therefore conclusions about longer-horizon gains are correspondingly more cautious.

4.2. Interpreting Embedding dimensions

4.3. Geometric Feature regressions

References

1. Andersen, T. G., & Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. International Economic Review, 39(4), 885–905.

2. Andersen, T. G., Bollerslev, T., Diebold, F. X., & Labys, P. (2003). Modeling and forecasting realized volatility. Econometrica, 71(2), 579–625.

3. Ball, R., & Brown, P. (1968). An empirical evaluation of accounting income numbers. Journal of Accounting Research, 6(2), 159–178.

4. Bernard, V. L., & Thomas, J. K. (1989). Post-earnings-announcement drift: Delayed price response or risk premium? Journal of Accounting Research, 27, 1–36.

5. Bernard, V. L., & Thomas, J. K. (1990). Evidence that stock prices do not fully reflect the implications of current earnings for future earnings. Journal of Accounting and Economics, 13(4), 305–340.

6. Chen, J. V., Nagar, V., & Schoenfeld, J. (2018). Manager–analyst conversations in earnings conference calls. Review of Accounting Studies, 23(4), 1315–1354.

7. Corsi, F. (2009). A simple approximate long-memory model of realized volatility. Journal of Financial Econometrics, 7(2), 174–196.

8. Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. Journal of Business & Economic Statistics, 13(3), 253–263.

9. Frankel, R., Johnson, M., & Skinner, D. J. (1999). An empirical examination of conference calls as a voluntary disclosure medium. Journal of Accounting Research, 37(1), 133–150.

10. Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. Annals of statistics, 1189-1232.

11. Gentzkow, M., Kelly, B., & Taddy, M. (2019). Text as data. Journal of Economic Literature, 57(3), 535–574.

12. Jolliffe, I. T. (2002). Principal Component Analysis (2nd ed.). Springer.

13. Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... & Liu, T. Y. (2017). Lightgbm: A highly efficient gradient boosting decision tree. *Advances in neural information processing systems*, *30*.

14. Ledoit, O., & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.

15. Ljung, G. M., & Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.

16. Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. Journal of Finance, 66(1), 35–65.

17. Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*.

18. Lundberg, S. M., Erion, G., & Lee, S.-I. (2020). Consistent individualized feature attribution for tree ensembles. arXiv preprint arXiv:1802.03888.

19. Mahalanobis, P. C. (1936). On the generalised distance in statistics. Proceedings of the National Institute of Sciences of India, 2(1), 49–55.

20. Matsumoto, D., Pronk, M., & Roelofsen, E. (2011). What makes conference calls useful? The information content of managers' presentations and analysts' discussion sessions. The Accounting Review, 86(4), 1383–1414.

21. Newey, W. K., & West, K. D. (1986). A simple, positive semi-definite, heteroskedasticity and autocorrelationconsistent covariance matrix.

22. Price, S. M., Doran, J. S., Peterson, D. R., & Bliss, B. A. (2012). Earnings conference calls and stock returns: The incremental informativeness of textual tone. (Working paper; circulated under this title).

23. Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. arXiv preprint arXiv:1908.10084.

24. Shapley, L. S. (1953). A value for n-person games. In H. W. Kuhn & A. W. Tucker (Eds.), Contributions to the Theory of Games II (pp. 307–317). Princeton University Press.

25. Tasker, S. C. (1998). Bridging the information gap: Quarterly conference calls as a medium for voluntary disclosure. Review of Accounting Studies, 3, 137–167.

Appendix

- OLS vs LGBM baseline HAR
- Shock validation (conditional on length)